

Paper I v1.3 Package-Residual Internal Audit

Internal author-side audit — not external

21 August 2026 — Overall verdict: PASS

Protocol: PAPER_I_INTERNAL_PACKAGE_RESIDUAL_v1.0.

Lean execution: no. **Manuscript/PDF regeneration:** no; all six artifacts are byte-identical.

1 Target

The audit covers the same Paper I `preprint_draft_v1.3` manuscript and formal artifacts, with two package-only corrections after the external `PASS_WITH_RESIDUALS` report.

Anchor	SHA-256
English Markdown	f3094b670c93ff622c3f573cdab61bdd0f5d84007f04b1888b364e0183565bea
Spanish Markdown	57ba967fc2de805b7bbb4cf5f937727bde43c2ada80fa794bcbb5a727db05b8b
English PDF	7d04c47692b613c8d6e2cc4471f0205128b4ae06ca11d581a08d020eb2236db0
Spanish PDF	3ad16ae86e8fcad358bf964a6ae98ae053b0bde2fdf3140e008cedab78b90c2a
Lean archive	0181506408644fc1f8872d711de5a98a500f4052aa295bcd6f8c82776694fd3a
External report	f2ad1605f0a802932c07503bfad429a98b08af26844dd968aad6e3f145aee495

The package-fix input freeze contains 219 files and 24,495,011 bytes. Its aggregate path/hash-list SHA-256 is 402fe23d5538851b290c749927ea75052efadc8c5df5cda5aba5fdbe0495a325.

2 Corrections

RES-V13-001 — PASS. The active package no longer contains `tmp/internal_report_v1.3/`. No `tmp` directory, forbidden TeX scratch extension, zero-byte file or stray `$o` occurs in the frozen target.

RES-V13-002 — PASS. The changelog now names `PaperI.assembly_sharp` and `PaperI.Split.residual_duality`. Each occurs exactly once; the transposed forms are absent. The corrected names agree with Appendix C, `FreezeAxioms.lean` and the recorded axiom output.

RES-V13-003 was already closed by external primary-source verification. RES-V13-004 remains a nonblocking NOTE about a third-party certificate; it is not represented as repaired.

3 Gate results

Gate	Verdict	Evidence
R0	PASS after harness correction	20/20 checks; both open MINOR findings closed.
R1	PASS	95/95 canonical static checks.
R2	PASS	5 identities; 561 boundary; 480 orbit; 99,671 assembly; 1,179 split-LP; 9 sharpness instances.
R3	PASS	Zero exact/near duplicate blocks across all six artifacts.
R4	PASS recorded/external reuse	Unchanged archive; 8,034-job build and 15-surface axiom gate; no Lean rerun.
R5	PASS unchanged artifacts	Unchanged hashes; 19/20 pages, embedded fonts, 39 fresh renders inspected.
R6	PASS	Gate manifests, archive and sidecar generated last and verified.

4 Diagnostic transparency

The first R0 run used the wrong base for `CURRENT_TARGET_SHA256.txt`; the harness was corrected and rerun. An attempted nested execution of R1 also derived the wrong repository root; the canonical script was then used. Neither diagnostic changed the target. Both final runs pass.

5 Semantic integrity

No manuscript, theorem, definition, assumption, quantifier, constant, equation, proof step, citation claim, figure, LaTeX source, PDF or Lean source changed. Rebuilding unchanged PDFs or Lean would add operational risk without testing either corrected defect.

6 Verdict

PASS. All seven residual internal gates pass. There are zero unresolved blockers, majors or minors from the external report. This author-side result is not human peer review and does not prove global novelty. A final external residual re-audit must issue its own plain **PASS** before Paper I is externally closed.